Euclid's Elements — Book I
Proposition 22

CGJteamLab

To construct a triangle out of three straight lines which equal three given straight lines: thus it is necessary that two of the straight lines taken together in any manner should be greater than the remaining one.

1 Introduction

Euclid's Proposition I.22 is the classical existence theorem for a triangle with three prescribed side lengths.

Three nondegenerate segments are given. Euclid assumes that any two of them, taken together, are greater than the remaining one. Under these hypotheses he constructs a triangle whose three sides are congruent to the three given segments.

In the present reconstruction the given segments are denoted

$$Aa, \qquad Bb, \qquad Cc.$$

The three hypotheses are the strict triangle inequalities

$$Aa < Bb + Cc, \qquad Bb < Aa + Cc, \qquad Cc < Aa + Bb.$$

The formal conclusion is the existence of a point K such that

$$BK \cong Aa \qquad \text{and} \qquad bK \cong Cc.$$

Since Bb is already the third prescribed side, the triangle BbK has precisely the required side lengths.

Proposition I.22 is especially important for the formal development because its proof exposes two pieces of infrastructure which classical notation tends to hide:

1. the phrase "two segments taken together are greater than a third" must receive a synthetic, witness-based meaning;
2. the existence of the vertex K is an application of circle-circle continuity.

No numerical length function is introduced. Segment sums, strict inequalities, and the circle intersection are all expressed in the synthetic language of betweenness and congruence.

The resulting formal architecture is therefore different from the visual shortcut suggested by the classical drawing:

```
triangle inequalities
    |
    +-- synthetic sum witnesses
    |
    +-- construct a point on the G-circle
    |       that lies inside the F-circle
    |
    +-- construct a point on the G-circle
    |       that lies outside the F-circle
    |
    v
circle-circle continuity
    |
    v
intersection point K
    |
    v
BK ~= Aa,  bK ~= Cc
```

This is the hidden logical content of the apparently simple instruction "draw two intersecting circles."

2 The Classical Configuration

Euclid's construction is best understood in the classical ray configuration used by Wilkins. Let the three prescribed lengths be represented by A, B, and C. On one ray choose consecutive segments

$$DF = A, \qquad FG = B, \qquad GH = C.$$

Draw the circle centered at F with radius A, and the circle centered at G with radius C. Under the three strict triangle inequalities the circles intersect properly in two points K and L. Either triangle FGK or FGL then has side lengths A, B, C.

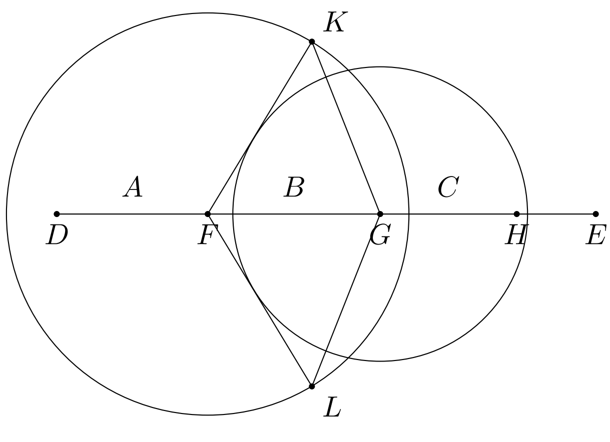

Figure 1: Wilkins-style configuration for Euclid I.22. The circles centered at F and G intersect in K and L, producing triangles with side lengths A, B, C.

The important point is that the three inequalities are not decorative side conditions. They control the relative position of the two circles. Wilkins makes this explicit by displaying the six boundary and failure configurations obtained when one of the strict triangle inequalities is replaced by equality or reversed.

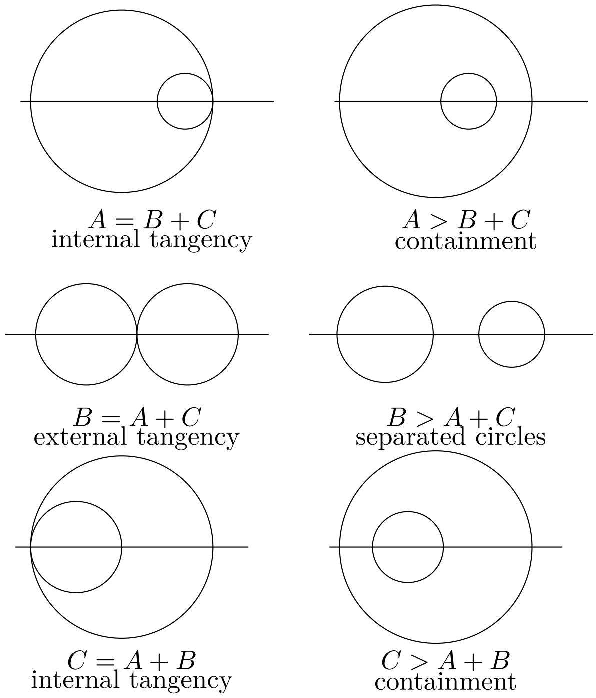

Figure 2: The six boundary and failure configurations corresponding to the three triangle inequalities. Equality gives tangency; reversing the inequality destroys proper intersection.

This makes the geometric meaning of the hypotheses visible:

```
A < B + C
B < A + C
C < A + B
      |
      v
two circles intersect properly
      |
      v
triangle with sides A, B, C
```

The formal proof does not use a numerical metric criterion for circle intersection. Instead, it constructs explicit witnesses on one circle that are respectively inside and outside the other circle. Circle-circle continuity then produces the common point.

3 Classical Proof

Euclid's proof is a direct ruler-and-compass construction.

Let the three prescribed segments be represented by

$$Aa, \qquad Bb, \qquad Cc,$$

and assume that every two taken together are greater than the remaining one:

$$Aa < Bb + Cc, \qquad Bb < Aa + Cc, \qquad Cc < Aa + Bb.$$

Choose the segment Bb as the base of the required triangle. Draw

- the circle centered at B with radius congruent to Aa;
- the circle centered at b with radius congruent to Cc.

Let K be an intersection point of these two circles. By construction,

$$BK \cong Aa \qquad \text{and} \qquad bK \cong Cc.$$

Together with the base Bb, the triangle BbK therefore has the three prescribed side lengths.

The classical proof may be summarized as

$$\begin{gathered} Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb \\ \downarrow \\ \text{draw two circles with centers } B \text{ and } b \\ \downarrow \\ K \text{ is their intersection} \\ \downarrow \\ BK \cong Aa, \qquad bK \cong Cc \\ \downarrow \\ \text{triangle } BbK. \end{gathered}$$

The point at which the axiomatic reconstruction must say more is the existence of K. In the traditional diagram the two circles are simply drawn as intersecting. A formal proof must derive an appropriate intersection theorem from explicit hypotheses.

Thus Proposition I.22 is one of the places where the difference between a construction diagram and an axiomatic existence proof becomes especially visible.

4 Statement

The synthetic statement used in the final Lean theorem is:

```
theorem euclid_proposition_22
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A a B b C c : Geo.Point)
    (hAa : A != a)
    (hBb : B != b)
    (hCc : C != c)
    (hAa_lt_BbCc :
      HilbertSegmentSumGreater Geo B b C c A a)
    (hBb_lt_AaCc :
      HilbertSegmentSumGreater Geo A a C c B b)
```

```
    (hCc_lt_AaBb :
      HilbertSegmentSumGreater Geo A a B b C c) :
    exists K : Geo.Point,
      Geo.Congruent B K A a /\
      Geo.Congruent b K C c
```

The three assumptions involving `HilbertSegmentSumGreater` express

$$Bb + Cc > Aa, \qquad Aa + Cc > Bb, \qquad Aa + Bb > Cc.$$

The conclusion gives the third vertex K of the required triangle:

$$BK \cong Aa, \qquad bK \cong Cc,$$

while the base itself is the given segment Bb.

5 A Synthetic Predicate for "Two Together Greater"

Proposition I.21 showed that segment sums need not in general be reified as numerical objects. A concrete collinear segment can serve as a representative of a sum.

Proposition I.22 introduces a slightly different requirement. The three triangle inequalities are not merely intermediate steps in the proof: they are part of the public hypotheses of the theorem. Consequently it is useful to package the geometric witness explicitly.

The Book Zero predicate is

```
def HilbertSegmentSumGreater
    [HilbertCongruence Geo]
    (A B C D E F : Geo.Point) : Prop :=
  exists P : Geo.Point,
    Geo.Between A B P /\
    Geo.Congruent B P C D /\
    HilbertSegmentLess Geo E F A P
```

Geometrically, a witness P satisfies

$$A - B - P, \qquad BP \cong CD.$$

Therefore AP represents the sum

$$AB + CD.$$

The final condition

$$EF < AP$$

says precisely

$$AB + CD > EF.$$

For example,

`HilbertSegmentSumGreater Geo B b C c A a`

means that there exists P with

$$B - b - P, \qquad bP \cong Cc, \qquad Aa < BP.$$

Thus BP is a synthetic representative of $Bb + Cc$.

This definition does not introduce an algebra of lengths. It merely packages an ordinary geometric construction together with the strict comparison already provided by `HilbertSegmentLess`.

6 Proof Architecture of the Reconstruction

The reconstruction is intentionally layered. The final theorem is short because the difficult circle geometry is isolated in reusable helpers.

The overall dependency is

$$\begin{array}{c} \text{three triangle inequalities} \\ \downarrow \\ \texttt{HilbertSegmentSumGreater} \\ \downarrow \\ P, Q, R \text{ representing the three sums} \\ \downarrow \\ \texttt{euclid\_proposition\_22\_circle\_witnesses} \\ \downarrow \\ N \text{ inside}, \quad H \text{ outside} \\ \downarrow \\ \texttt{euclid\_proposition\_22\_circle\_intersection} \\ \downarrow \\ K \\ \downarrow \\ BK \cong Aa, \quad bK \cong Cc. \end{array}$$

The principal intermediate results are discussed below.

6.1 Transporting the outside inequality

The helper

`euclid_proposition_22_outside_helper`

handles the synthetic arithmetic behind the outside point.

Suppose BP represents

$$Bb + Cc,$$

and another collinear segment FH is assembled from pieces

$$FG \cong Bb, \qquad GH \cong Cc.$$

Using `bookZero_sumOfParts`, the proof obtains

$$BP \cong FH.$$

Hence the triangle inequality

$$Aa < BP$$

can be transported to

$$Aa < FH$$

by `bookZero_30_lessThanCongruence`.

This is the formal replacement for the classical step "equal sums may be substituted in an inequality."

6.2 Same-ray trichotomy

A point constructed on a ray can lie before, at, or beyond the reference point. The helper

`euclid_proposition_22_sameRay_trichotomy`

makes this order distinction explicit.

For points G, F, N on the same ray from G, it proves

$$G - N - F \quad \vee \quad N = F \quad \vee \quad G - F - N.$$

The proof uses strict segment trichotomy together with `bookZero_51_lessThanBetween` and layoff uniqueness.

This case distinction is essential when the point N on the second circle is compared with the first circle.

6.3 Cancellation for strict segment inequalities

The helper

`euclid_proposition_22_lessThan_cancel_right`

is a synthetic cancellation principle.

If

$$A - B - E, \qquad C - D - F,$$

the terminal parts satisfy

$$BE \cong DF,$$

and

$$AE < CF,$$

then

$$AB < CD.$$

Classically this is an immediate cancellation of equal addends. Formally it requires a proof from the developed strict-order and congruence theory.

This helper is one of the clearest examples in I.22 of Book Zero providing arithmetic-like behavior without numerical lengths.

6.4 The inside point

The proof constructs a point N on the ray from b through B such that

$$bN \cong Cc.$$

The two possible order configurations are handled by

```
euclid_proposition_22_inside_left
euclid_proposition_22_inside_right
euclid_proposition_22_inside_helper
```

and the same-ray trichotomy theorem.

The goal is not merely to locate N on the second circle. One must also prove that it is inside, or on the center of, the first circle:

$$N = B \quad \vee \quad BN < Aa.$$

The relevant triangle inequalities are used to derive this comparison in each possible collinear configuration.

6.5 Constructing explicit sum representatives

The helper

```
euclid_proposition_22_extend_with_segment
```

constructs the ordinary geometric representative of a sum.

Starting from a nondegenerate segment AB and another segment CD, it extends the ray through B to a point P satisfying

$$A - B - P, \qquad BP \cong CD.$$

Hence

$$AP$$

represents

$$AB + CD.$$

This is the basic geometric operation behind the witnesses P, Q, R used in the core construction.

6.6 Circle witnesses

The theorem

```
euclid_proposition_22_circle_witnesses
```

is the main preparatory result.

From the three represented triangle inequalities it constructs points H and N on the circle centered at b with radius congruent to Cc, with the following properties:

$$bH \cong Cc, \qquad Aa < BH,$$

and

$$bN \cong Cc, \qquad N = B \ \vee \ BN < Aa.$$

Thus H is outside the circle centered at B with radius Aa, while N is inside it or coincides with its center.

This is precisely the configuration required by circle-circle continuity.

7 Circle-Circle Continuity

The central existence step is isolated in

`euclid_proposition_22_circle_intersection`

The Hilbert circle-continuity theorem does not use the radius Aa directly as a numerical quantity. The proof first lays off a congruent segment BT:

$$BT \cong Aa.$$

The first circle is therefore represented by center B and radius segment BT.

The second circle has center b and radius bN, with

$$bN \cong Cc.$$

The already constructed points satisfy

$$bH \cong bN,$$

so H lies on the second circle, and

$$Aa < BH.$$

After transport through $BT \cong Aa$, this becomes

$$BT < BH,$$

showing that H is outside the first circle.

Similarly,

$$N = B \quad \vee \quad BN < Aa$$

is transported to

$$N = B \quad \vee \quad BN < BT,$$

showing that N is inside the first circle.

The theorem

```
hilbert_circle_circle_intersection
```

then yields a point K satisfying

$$BK \cong BT \qquad \text{and} \qquad bK \cong bN.$$

Finally congruence transitivity returns to the original given segments:

$$BK \cong Aa, \qquad bK \cong Cc.$$

This is the required vertex.

8 The Core and the Public Theorem

Once the circle witnesses and the circle-intersection theorem are available, the remaining layers become very short.

The theorem

```
euclid_proposition_22_core
```

accepts explicit representatives P, Q, R of the three sums:

$$BP = Bb + Cc, \qquad AQ = Aa + Cc, \qquad AR = Aa + Bb,$$

together with

$$Aa < BP, \qquad Bb < AQ, \qquad Cc < AR.$$

It invokes the two successive helpers

```
euclid_proposition_22_circle_witnesses
euclid_proposition_22_circle_intersection
```

The intermediate wrapper

```
euclid_proposition_22_from_sum_witnesses
```

packages the same construction when all three sum witnesses are supplied explicitly.

The public theorem finally removes P, Q, R from its interface. It opens the three witnesses stored in `HilbertSegmentSumGreater` and passes them to the internal construction.

Thus the public API states exactly the geometric theorem rather than the implementation details of its proof.

9 Lean Formalization

The public theorem deliberately exposes only the three given segments, their nondegeneracy, the three triangle inequalities, and the required vertex:

```
theorem euclid_proposition_22
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A a B b C c : Geo.Point)
    (hAa : A != a)
    (hBb : B != b)
    (hCc : C != c)
    (hAa_lt_BbCc :
      HilbertSegmentSumGreater Geo B b C c A a)
    (hBb_lt_AaCc :
      HilbertSegmentSumGreater Geo A a C c B b)
    (hCc_lt_AaBb :
      HilbertSegmentSumGreater Geo A a B b C c) :
    exists K : Geo.Point,
      Geo.Congruent B K A a /\
      Geo.Congruent b K C c
```

The public assumptions are therefore the exact synthetic counterparts of

$$Aa < Bb + Cc, \qquad Bb < Aa + Cc, \qquad Cc < Aa + Bb.$$

Internally, each occurrence of `HilbertSegmentSumGreater` is unpacked into a concrete witness. Schematically:

```
rcases hAa_lt_BbCc with
  <P, hBbP, hbP_Cc, hAa_BP>

rcases hBb_lt_AaCc with
  <Q, hAaQ, haQ_Cc, hBb_AQ>

rcases hCc_lt_AaBb with
  <R, hAaR, haR_Bb, hCc_AR>
```

The points P, Q, R are not abstract numbers representing sums. They are ordinary points producing collinear segments whose lengths realize the three sums geometrically.

The next layer constructs the circle witnesses. In schematic form:

```
have hWitnesses :=
  euclid_proposition_22_circle_witnesses
    ...

rcases hWitnesses with
  <H, N, hH_on_circle, hH_outside,
         hN_on_circle, hN_inside>
```

The important output is the inside/outside configuration

$$Aa < BH, \qquad N = B \ \lor \ BN < Aa,$$

while both H and N lie on the circle centered at b with radius congruent to Cc.

The actual existence step is then concentrated in

`euclid_proposition_22_circle_intersection`

which invokes

`hilbert_circle_circle_intersection`

after transporting the radius and inequality data into the orientation expected by the continuity theorem.

Finally, congruence transitivity converts the constructed radius representatives back to the original segments and yields

$$BK \cong Aa, \qquad bK \cong Cc.$$

The Lean proof is therefore longer than Euclid's paper proof not because it changes the construction, but because it makes three suppressed layers explicit:

$$\text{segment arithmetic} \ + \ \text{ray/order geometry} \ + \ \text{circle continuity}.$$

10 Relationship with Earlier Propositions

10.1 Proposition I.20

Proposition I.20 established the triangle inequality inside an already given triangle.

Proposition I.22 uses the converse direction as an existence hypothesis: three prescribed segment lengths satisfying the three triangle inequalities should determine a triangle.

The logical roles are therefore different. I.20 proves inequalities from a triangle; I.22 assumes the inequalities in order to construct a triangle.

10.2 Proposition I.21

Proposition I.21 demonstrated that segment sums can be handled synthetically by constructing ordinary segments representing sums of parts.

I.22 develops this idea further. Because the triangle inequalities are part of the public statement rather than temporary intermediate facts, the witness-bearing predicate `HilbertSegmentSumGreater` is introduced as an interface notion.

Thus I.22 turns the segment-arithmetic techniques of I.21 into reusable public infrastructure.

10.3 Proposition I.23

Proposition I.23 follows I.22 in Euclid's sequence and uses it essentially in Euclid's own construction of an angle congruent to a given angle.

In the Hilbert reconstruction, however, I.23 does not formally depend on I.22 because Hilbert III.4 already supplies angle construction.

This makes I.22 a useful reference point for distinguishing the historical dependency graph of the *Elements* from the dependency graph induced by the Hilbert foundation.

11 Hilbert and Book Zero Components Used

11.1 HilbertSegmentLess

The relation

```
HilbertSegmentLess
```

is the synthetic strict order on segments.

It supplies the comparisons needed to distinguish points lying inside and outside a circle and supports the transport and transitivity operations used throughout the proof.

11.2 bookZero_49_layoff

The theorem

```
bookZero_49_layoff
```

copies a given nondegenerate segment onto a chosen ray.

In I.22 it is used to represent the radius Aa by a segment based at the center B:

$$BT \cong Aa.$$

11.3 bookZero_sumOfParts

The theorem

```
bookZero_sumOfParts
```

recognizes congruent wholes assembled from congruent adjacent parts.

It is used when a segment representing $Bb + Cc$ must be identified with another segment built from a copy of Bb followed by a copy of Cc.

11.4 Congruence transport for inequalities

The theorems

```
bookZero_30_lessThanCongruence
bookZero_32_lessThanCongruence2
```

transport strict segment comparisons through congruent representatives.

They replace the informal algebraic operation of substituting equal lengths into inequalities.

11.5 bookZero_31_trichotomy1

This theorem supplies the equality case when neither of two nondegenerate segments is strictly shorter than the other.

Together with layoff uniqueness it provides the middle case

$$N = F$$

in the same-ray trichotomy used by I.22.

11.6 bookZero_51_lessThanBetween

The theorem

```
bookZero_51_lessThanBetween
```

converts strict segment comparison on a common ray into strict betweenness.

It is the bridge between the order relation on segment lengths and the incidence-order geometry needed to locate the constructed circle points.

11.7 Circle-circle intersection

The theorem

```
hilbert_circle_circle_intersection
```

is the decisive existence principle.

Given a point of one circle lying inside a second circle and another point of the same first circle lying outside the second, it produces a common point of the two circles.

Proposition I.22 is therefore not merely an application of elementary segment arithmetic. It is the point where that arithmetic is used to establish the hypotheses of a genuine continuity theorem.

12 Comparison with Euclid's Original Organization

Euclid presents I.22 as an explicit ruler-and-compass construction. He places copies of the three given segments on a straight line and draws two circles whose intersection gives the required triangle.

The Hilbert reconstruction preserves the same geometric idea but changes its logical organization.

First, the base Bb is already available as one of the given segments, so there is no need to reproduce Euclid's preliminary line solely to manufacture a base of the correct length.

Second, the three assumptions "two together greater than the remaining one" are exposed through `HilbertSegmentSumGreater`. Their witnesses give concrete segments representing the required sums.

Third, the existence of the intersection is not treated as visually obvious. Explicit points N and H are constructed on one circle, and they are proved to lie respectively inside and outside the other. Only then is circle-circle continuity invoked.

Thus the formal proof separates three issues which Euclid's diagram compresses:

$$\text{segment arithmetic} \quad + \quad \text{order on rays} \quad + \quad \text{continuity.}$$

The resulting proof is longer, but the additional length identifies the exact logical content required for the construction.

13 What Proposition I.22 Adds to Book Zero

I.21 suggested that explicit segment-sum objects were unnecessary: inside a proof, sums could be represented directly by constructed segments.

I.22 refines that conclusion rather than contradicting it.

The new predicate

`HilbertSegmentSumGreater`

does not turn segment sums into algebraic objects. It is a witness-bearing proposition:

$$AB + CD > EF$$

means that a point exists which geometrically realizes $AB + CD$, and that the resulting whole segment is strictly greater than EF.

This distinction is important.

The project still has no numerical length map and no binary addition operation on lengths. What has been added is a convenient *interface proposition* for a classical phrase that occurs as a hypothesis of Euclid's theorem.

Proposition I.22 also shows that the Book Zero layer is now strong enough to prepare the exact inside/outside data required by a continuity theorem. The layer is therefore doing more than abbreviating elementary congruence arguments: it is connecting synthetic segment arithmetic with topological existence principles.

14 Dependency Summary

The public assumptions are

$$Aa \neq 0, \quad Bb \neq 0, \quad Cc \neq 0,$$

$$\begin{aligned} Aa &< Bb + Cc, \\ Bb &< Aa + Cc, \\ Cc &< Aa + Bb. \end{aligned}$$

In Lean the three inequalities are encoded by

```
HilbertSegmentSumGreater Geo B b C c A a,
HilbertSegmentSumGreater Geo A a C c B b,
HilbertSegmentSumGreater Geo A a B b C c.
```

The complete proof architecture is

$$\begin{gathered}
\text{triangle inequalities} \\
\downarrow \\
P, Q, R \text{ sum witnesses} \\
\downarrow \\
\text{segment transport} + \text{same-ray order} \\
\downarrow \\
N = B \ \vee \ BN < Aa \qquad Aa < BH \\
\downarrow \\
bN \cong Cc, \qquad bH \cong Cc \\
\downarrow \\
\texttt{hilbert_circle_circle_intersection} \\
\downarrow \\
BK \cong Aa, \qquad bK \cong Cc \\
\downarrow \\
\texttt{euclid_proposition_22}.
\end{gathered}$$

No new geometric axiom is introduced in Proposition I.22. The proof uses the already available circle-circle continuity theorem together with the developed Hilbert and Book Zero segment theory.

15 Formalization Notes

Several features of I.22 are easy to miss in an informal proof.

First, a circle radius is represented by an ordinary segment, not by a real number. Even when the intended radius is Aa, the circle-continuity theorem is applied using a constructed representative BT with

$$BT \cong Aa.$$

Second, "inside a circle" is expressed synthetically by a strict segment comparison such as

$$BN < BT,$$

with the center case $N = B$ handled separately.

Third, "outside a circle" is likewise a strict segment comparison:

$$BT < BH.$$

Fourth, the fact that N lies on the ray from b through B is not enough to determine its order relative to B. The proof requires an explicit three-way analysis of the possible positions.

Fifth, cancellation of equal terminal pieces from strict inequalities is not automatic. It is isolated as a theorem and proved from the existing segment-order machinery.

Finally, the public theorem is deliberately cleaner than the internal construction. The implementation works with many auxiliary points, but the theorem exposes only the three original segments, their three triangle inequalities, and the required vertex.

16 Conclusion

Proposition I.22 is a major structural milestone in the reconstruction of Book I.

Its classical content is elementary: three admissible lengths determine a constructible triangle. Formalizing that statement reveals, however, a precise interaction between three layers of synthetic geometry.

The first layer is segment arithmetic. Betweenness, congruence, strict segment comparison, addition of adjacent parts, cancellation, and transport are used to express the three triangle inequalities and to manipulate their witnesses.

The second layer is ray and order geometry. Constructed points must be located explicitly on rays, and their possible orders must be distinguished.

The third layer is continuity. Once one point of the second circle is proved inside the first and another outside it, `hilbert_circle_circle_intersection` supplies the required common point.

The final theorem therefore has a particularly clear conceptual form: the triangle inequalities are exactly the hypotheses needed to build the inside/outside witnesses for two circles.

No coordinates, numerical lengths, or angle measures enter the proof. The reconstruction remains entirely synthetic, and the complete development through Euclid I.22 compiles successfully.